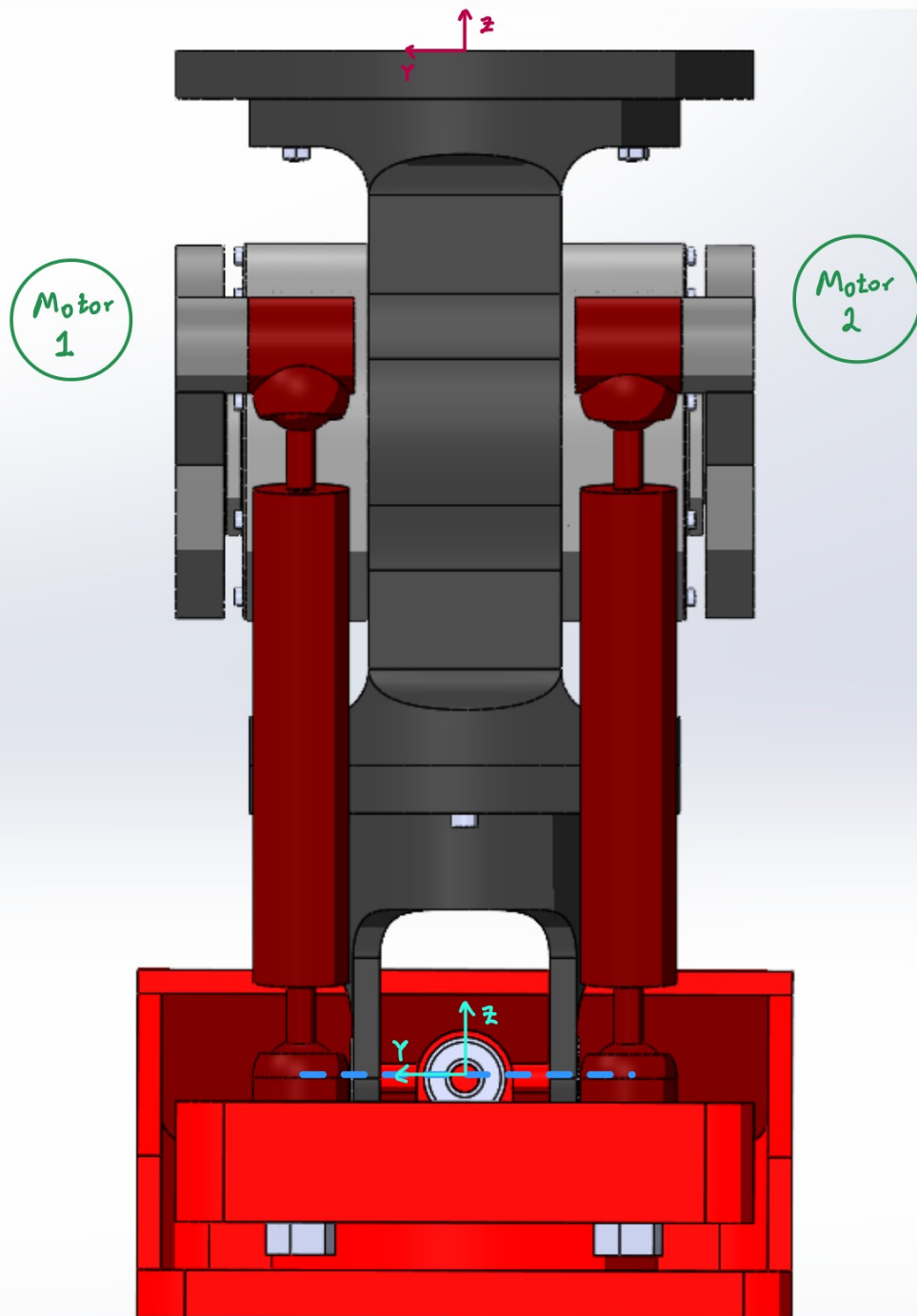


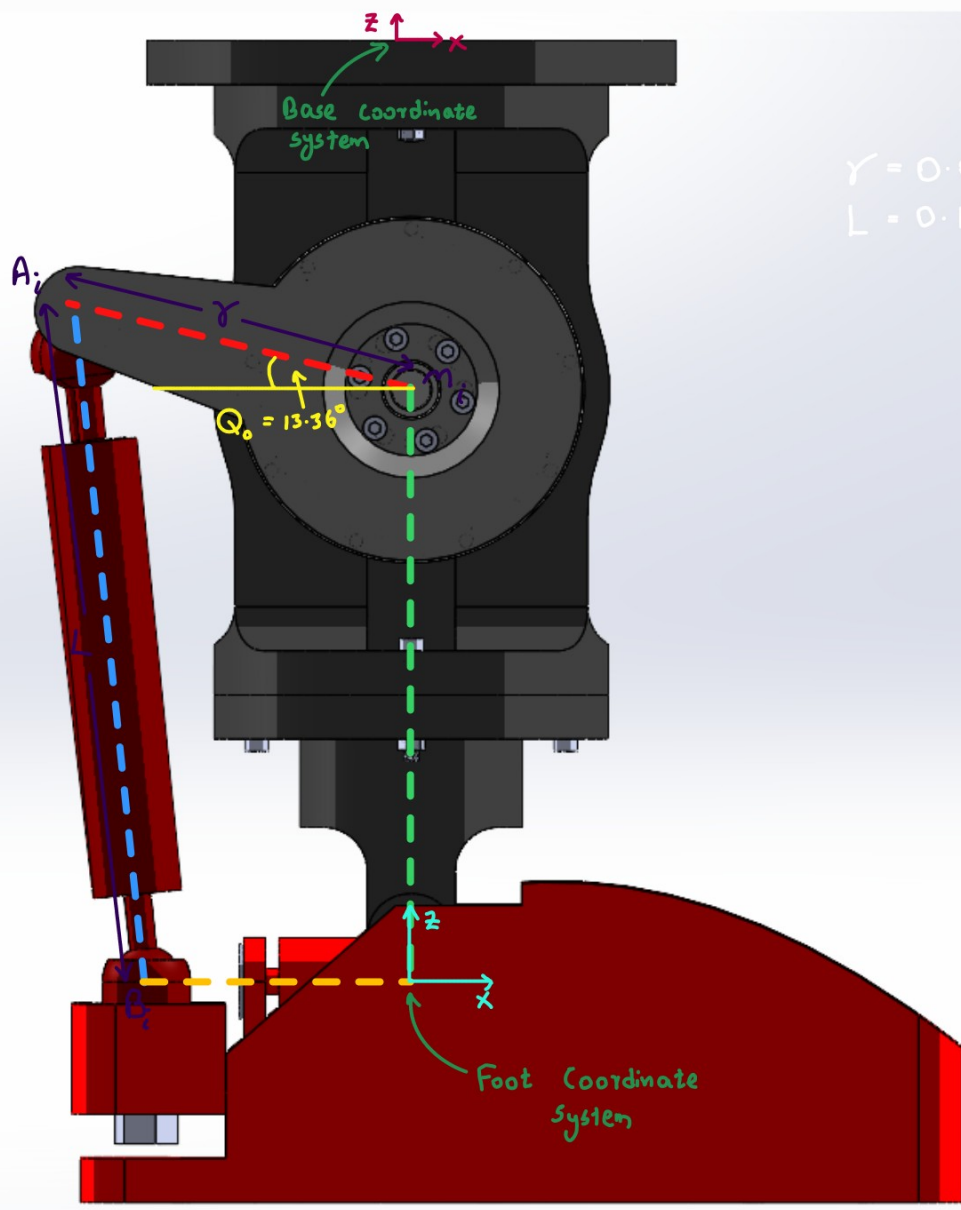


Motor positions:

$$\left. \begin{aligned} M_1 &\equiv [0, 0.06, -0.07925] \\ M_2 &\equiv [0, -0.06, -0.07925] \end{aligned} \right\} \text{w.r.t. Base coordinate system}$$

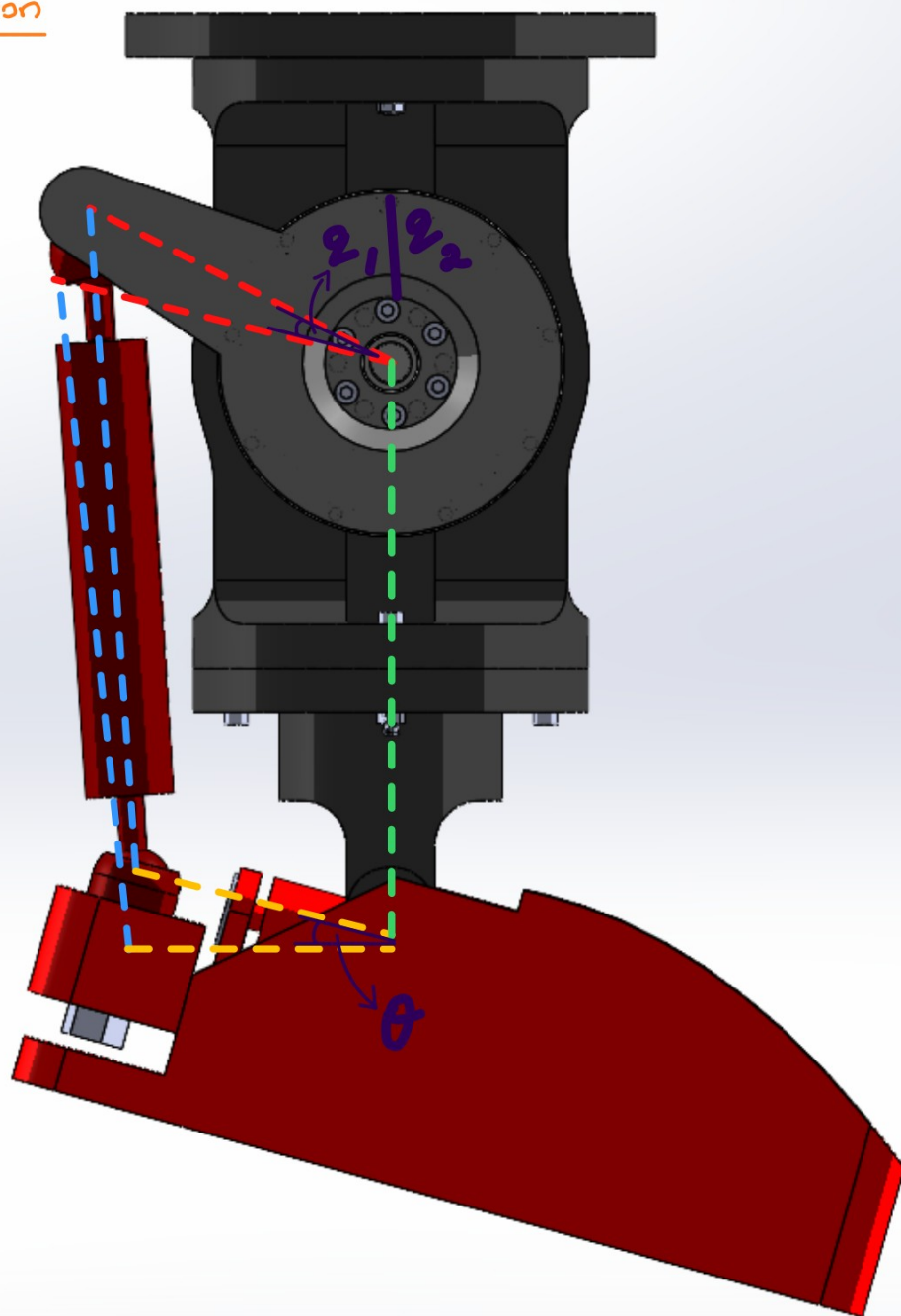
Foot-rod joint positions:

$$\left. \begin{aligned} P_1 &\equiv [-0.05304, 0.03006, 0] \\ P_2 &\equiv [-0.05304, -0.03006, 0] \end{aligned} \right\} \text{w.r.t. Foot coordinate system}$$

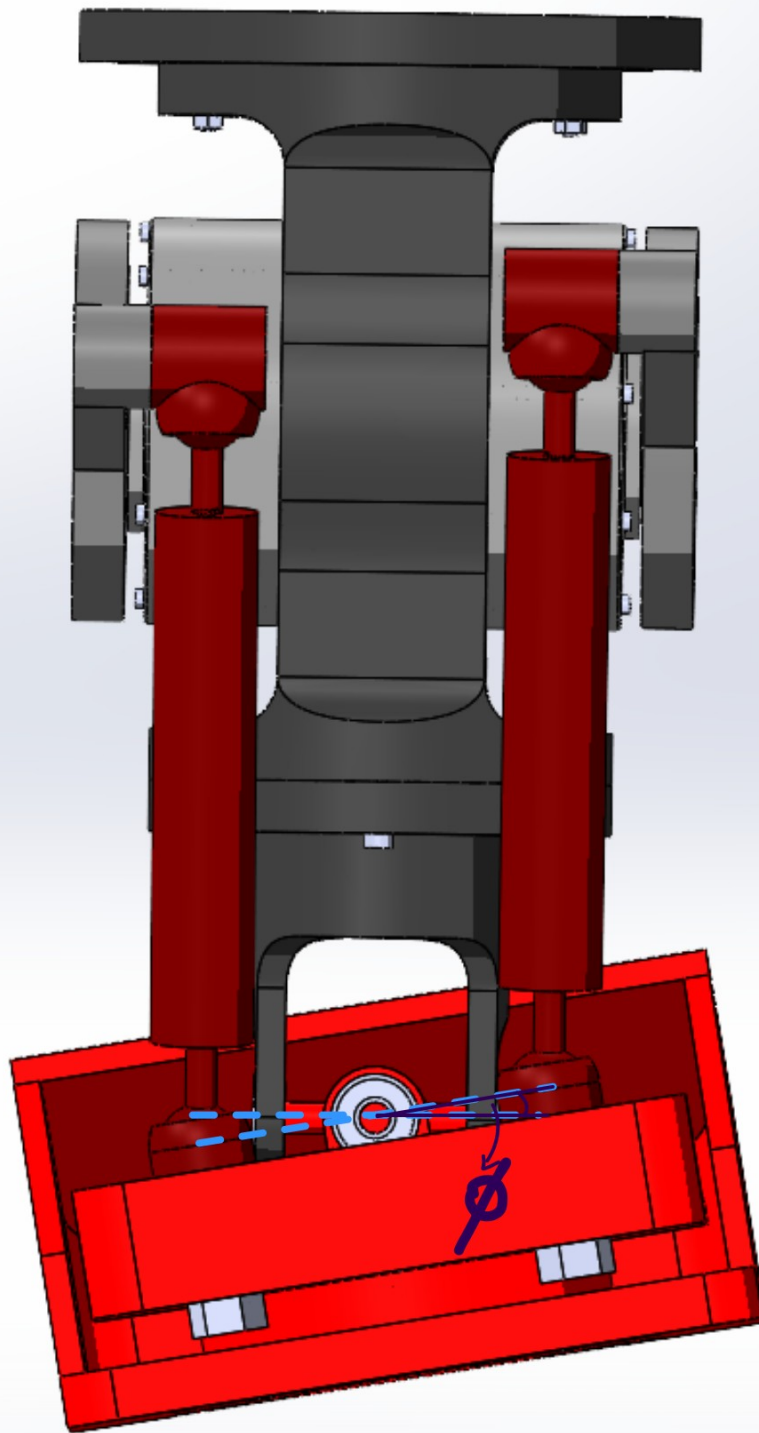


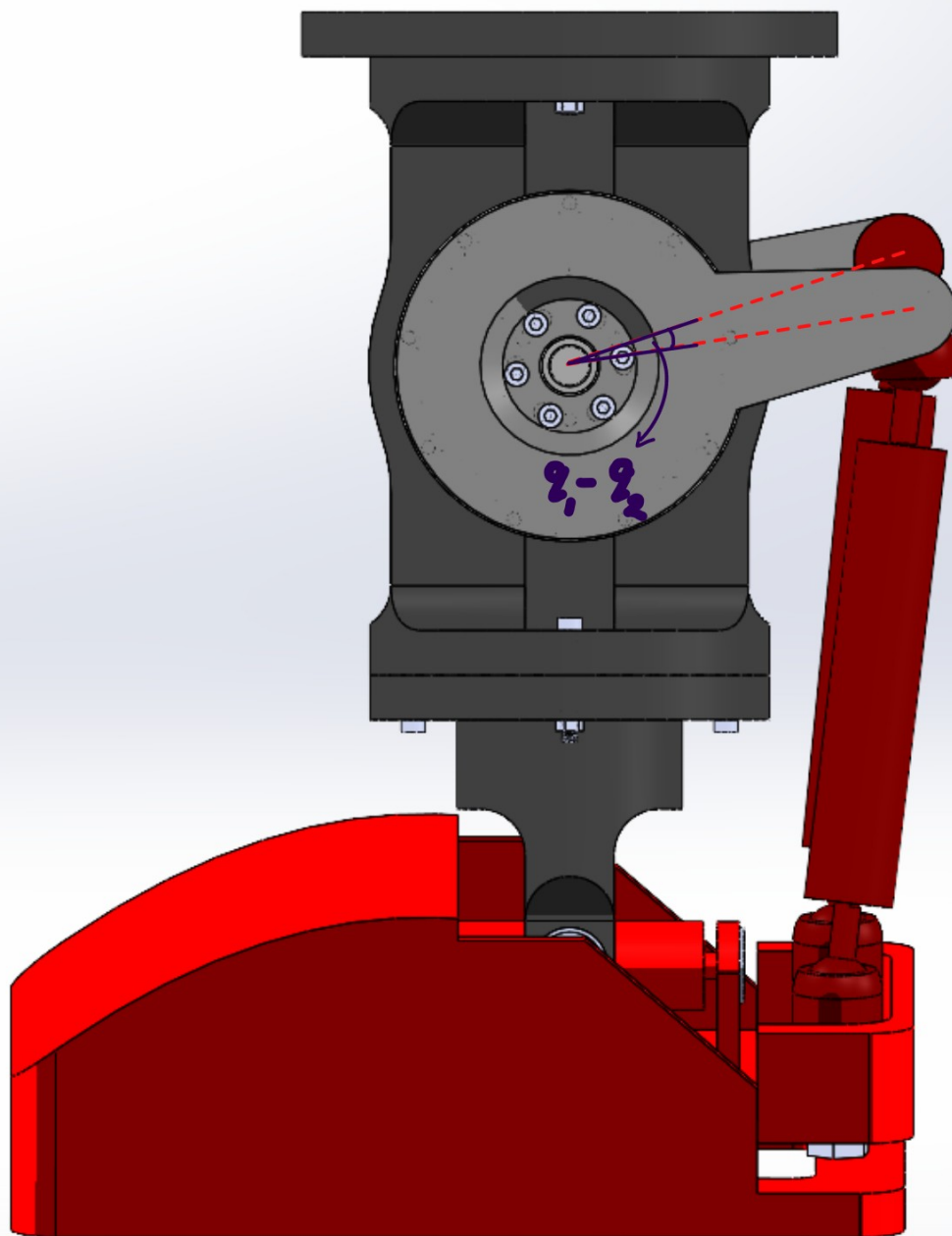
$$r = 0.0775 \text{ m}$$
$$L = 0.14315 \text{ m}$$

Pitch Motion



Roll Motion





$$B_i = R(\theta, \phi) \cdot P_i$$

$$R(\theta, \phi) = R_x(\phi) \cdot R_y(\theta)$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$

* Define,

$$d_i = B_i - M_i$$

$$d_i = \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$

4. Since motor crank always rotates in $x-z$ plane,

$$A_i = M_i + r \begin{bmatrix} \cos(q_i + Q_0) \\ 0 \\ \sin(q_i + Q_0) \end{bmatrix}$$

$$= \begin{bmatrix} M_{ix} + r \cos(q_i + Q_0) \\ M_{iy} \\ M_{iz} + r \sin(q_i + Q_0) \end{bmatrix}$$

5. Since the Rod has a fixed length,

$$|A_i - B_i| = L$$

$$(A_i - B_i)^T (A_i - B_i) = L^2$$

• From previous calculation,

$$B_i = d_i + M_i$$

$$\therefore A_i - B_i = A_i - d_i - M_i$$

$$= \begin{bmatrix} M_{ix} + r \cos(q_i + Q_0) \\ M_{iy} \\ M_{iz} + r \sin(q_i + Q_0) \end{bmatrix} - \begin{bmatrix} d_{ix} \\ d_{iy} \\ d_{iz} \end{bmatrix} - \begin{bmatrix} M_{ix} \\ M_{iy} \\ M_{iz} \end{bmatrix}$$

$$= \begin{bmatrix} r \cos(q_i + Q_0) - d_{ix} \\ -d_{iy} \\ r \sin(q_i + Q_0) - d_{iz} \end{bmatrix}$$

$$\therefore \begin{bmatrix} r \cos(q_i + Q_0) - d_{ix} \\ -d_{iy} \\ r \sin(q_i + Q_0) - d_{iz} \end{bmatrix}^T \begin{bmatrix} r \cos(q_i + Q_0) - d_{ix} \\ -d_{iy} \\ r \sin(q_i + Q_0) - d_{iz} \end{bmatrix} = L^2$$

$$[r \cos(q_i + Q_0) - d_{ix}]^2 + d_{iy}^2 + [r \sin(q_i + Q_0) - d_{iz}]^2 = L^2$$

$$r^2 \cos^2(q_i + Q_0) - 2r \cos(q_i + Q_0) d_{ix} + d_{ix}^2 + d_{iy}^2 + r^2 \sin^2(q_i + Q_0) - 2r \sin(q_i + Q_0) d_{iz} + d_{iz}^2 = L^2$$

$$r^2 + |d_i|^2 - 2r [\cos(q_i + Q_0) d_{ix} + \sin(q_i + Q_0) d_{iz}] = L^2$$

$$\therefore d_{ix} \cos(q_i + Q_0) + d_{iz} \sin(q_i + Q_0) = \frac{r^2 + |d_i|^2 - L^2}{2r}$$

* Let,

$$K = \frac{r^2 + |d_i|^2 - L^2}{2r}$$

then,

$$d_{ix} \cos(q_i + Q_0) + d_{iz} \sin(q_i + Q_0) = K$$

* Let,

$$\rho_i = \sqrt{d_{ix}^2 + d_{iz}^2}$$

$$\alpha_i = \text{atan2}(d_{iz}, d_{ix})$$

$$\Rightarrow d_{ix} = \rho_i \cos \alpha_i$$

$$d_{iz} = \rho_i \sin \alpha_i$$

$$\therefore \rho_i \cos \alpha_i \cos(q_i + Q_0) + \rho_i \sin \alpha_i \sin(q_i + Q_0) = K$$

$$\rho_i \cos[\alpha_i - (q_i + Q_0)] = K$$

$$\cos[\alpha_i - (q_i + Q_0)] = \frac{K}{\rho_i}$$

$$\therefore q_i = \alpha_i \pm \cos^{-1}\left[\frac{K}{\rho_i}\right] - Q_0$$

Elbow down / Elbow up

* Inverse kinematics equation,

↳ Mechanism only supports elbow up

$$q_i = \alpha_i + \cos^{-1}\left[\frac{K}{\rho_i}\right] - Q_0$$